

Comparative Discussion: KaNet5 vs. Standard-LeNet5

Although KaNet5 and Standard-LeNet5 achieve virtually identical **accuracy** and **loss** curves over 50 epochs, a multi-view analysis of their activation patterns (feature maps and Grad-CAM) suggests distinct encoding strategies.

A key observation is that KaNet5 routinely produces a substantial fraction of *negative* activations, driving its mean raw score close to zero and yielding large percentage deltas when compared to LeNet5's strictly positive activations. We hypothesize that this reflects KaNet5's intrinsic selectivity balancing excitatory and inhibitory responses to generate richer, more nuanced feature representations rather than indicating any deficiency.

- **Normalized feature-map views** (min max interval, absolute ranking, z-score + min max) tend to favor KaNet5 in terms of spatial concentration and signal-to-noise ratio. After scaling, KaNet5 appears to focus its activations on discriminative edges, corners, loops, and cavities, whereas LeNet5's high raw amplitudes may spread into noisier patterns.
- **Grad-CAM analyses** indicate that in the deepest layers KaNet5 produces cleaner saliency maps (deltas up to 43% on loops, 30% on edges and corners), which could imply superior high-level prototype formation and reduced structural noise.
- **Computational and energy implications:** KaNet5's mix of positive and negative activations may lead to *sparser* patterns and mutual cancellation, potentially lowering switching activity in MAC units and reducing dynamic power consumption. In contrast, LeNet5's uniformly positive activations might incur higher utilization and signal redundancy, suggesting that further regularization (batch normalization, weight pruning, spectral constraints) could be beneficial.

Despite these internal differences, both architectures attain similar performance metrics (accuracy and loss). This may be because final predictive accuracy relies chiefly on the ability to extract sufficiently discriminative *high-level* features rather than on raw activation amplitude or sparsity alone. LeNet5's abundant activations still capture enough informative patterns to

reach high accuracy, while KaNet5’s fewer but more selective activations achieve comparable outcomes.

Proposed hypotheses for further evaluation:

1. KaNet5’s negative activations reflect a form of built-in regularization that could reduce over-activation and improve generalization under distributional shifts.
2. The observed sparsity in KaNet5 may translate into measurable reductions in dynamic power consumption on hardware accelerators, warranting dedicated energy profiling.
3. LeNet5 could benefit substantially from explicit regularization techniques such as batch or layer normalization and structured pruning to mitigate redundant activations without sacrificing accuracy.
4. Task-specific evaluations (e.g., few-shot learning, anomaly detection, or robustness to noise) could verify whether KaNet5’s selective preservation of relevant features translates into practical performance advantages over LeNet5.
5. Further experiments with alternative Kan-based encoder architectures could explore whether preserving semantically relevant features while maintaining accuracy leads to improved overall performance in various application scenarios.

Normalization and Statistical Testing

Normalization Variants. Each raw heat-map H is transformed via one of four TRANSFORMS in `batch_scorer.py`:

- **raw**: identity, $H_{\text{raw}} = H$.
- **norm**: min max to $[0, 1]$:

$$H_{\text{norm}} = \frac{H - \min H}{\max H - \min H}.$$

- **abs_ordinal**: rank-order of $|H_{ij}|$, replacing each entry by its rank in the flattened absolute-value array.

- **zscore:**

1. compute $A = |H|$, then $Z = (A - \mu)/\sigma$;
2. scale Z to $[0, 1]$ via $(Z - \min Z)/(\max Z - \min Z)$.

Delta at Epoch 50. For each metric M and normalization, we compute the mean scores A_{50} (LeNet5) and B_{50} (KaNet5) at epoch 50 and report

$$\Delta@50 = 100 \frac{A_{50} - B_{50}}{|B_{50}|}.$$

E.g. for HeatxEdge under `abs_ordinal`, $\Delta@50 = 5.57\%$ (LeNet5 > KaNet5).

Δ_{range} (Range of Percentage Delta):

is defined as

$$\Delta_{\text{range}} = \max_{e=1,\dots,50} \Delta(e) - \min_{e=1,\dots,50} \Delta(e),$$

This metric measures the total swing in relative performance over training:

- A *small* Δ_{range} implies a *stable* advantage (the gap remains roughly the same at all epochs).
- A *large* Δ_{range} indicates that one models lead *fluctuates significantly* (e.g. early epochs favor one network, later epochs favor the other).

Sign. (Significance): The **Sign.** column records the result of the paired t -test with greater alternative via the binary search algorithm in `paired_ttest`. We label it

$$\text{accepted} \quad \text{if } 0.01 \leq p \leq 0.05,$$

Otherwise, we mark it **not informative** (either $p < 0.01$ or $p > 0.05$, or insufficient samples), indicating no reliable claim of one model systematically outperforming the other.

Changed: The **Changed** flag is set to **T** if and only if $\Delta(e)$ changes sign at least once over the 50 epochs (i.e. there exists e_1, e_2 such that $\Delta(e_1) > 0$ and $\Delta(e_2) < 0$). This signals a regime shift in which LeNet5 leads during some epochs and KaNet5 leads during others. If $\Delta(e)$ retains the same sign throughout, **Changed** is **F**, indicating a consistent direction of the effect.

Paired t -Test and Iterative p -Value Search. Significance is assessed by `paired_ttest`, which for the greater hypothesis performs a binary search over a scale factor f :

- Initialize `low` = -100 , `high` = 100 , `sign` = 1 .
- Repeat up to 100 iterations:
 1. Set `mid` = $(\text{low} + \text{high})/2$, $f = 1 + \text{sign} \times \text{mid}$.
 2. Compute

$$t = \frac{\bar{a} - f\bar{b}}{\sqrt{(\text{Var}(a) + f^2 \text{Var}(b) - 2f \text{Cov}(a, b))/N}}, \quad p = 1 - F_t(t; N - 1).$$

3. If $0.01 \leq p \leq 0.05$, stop; if $p > P_{\text{upper}}$, set `high` = `mid`, else `low` = `mid`.
- Return the final p and percent delta $100(f - 1)$.
- We label the test **accepted** when $0.01 \leq p \leq 0.05$, otherwise **not informative**.

Note on raw scores. Because each pixel-wise score is averaged over the map, **raw** activations yield extremely large $\Delta@50$ values whenever one model's mean activation is driven toward zero. In practice, KaNet5 produces many more negative pixel values in feature maps than LeNet5, so its mean raw score at epoch 50 is close to zero while LeNet5's mean remains strictly positive. Since

$$\Delta@50 = 100 \frac{A_{50} - B_{50}}{|B_{50}|},$$

and $|B_{50}|$ is very small for KaNet5, the percentage delta saturates to a very large number under **raw** normalization.

Interpretation of Normalizations

Each raw heat map H is transformed by one of the four **TRANSFORMS** in `batch_scorer.py`. From an XAI perspective each variant highlights different aspects of the models internal attention:

Table 1: Feature-map Metrics: Standard-LeNet5 vs. KaNet5

Metric	Norm. type	$\Delta@50$ (%)	Comparison@50	Δ_{range}	Sign.	Changed
heatxedge	raw	1406.25	LeNet5 > KaNet5 by 1406.25%	5523.68	accepted	F
heatxedge	norm	-9.92	KaNet5 > LeNet5 by 9.92%	5.95	accepted	F
heatxedge	abs_ordinal	5.57	LeNet5 > KaNet5 by 5.57%	3.43	accepted	F
heatxedge	zscore	53.10	LeNet5 > KaNet5 by 53.10%	19.38	accepted	F
heatxcorner	raw	1718.75	LeNet5 > KaNet5 by 1718.75%	9891.36	accepted	F
heatxcorner	norm	-12.97	KaNet5 > LeNet5 by 12.97%	6.71	accepted	F
heatxcorner	abs_ordinal	1.68	LeNet5 > KaNet5 by 1.68%	3.81	accepted	F
heatxcorner	zscore	46.69	LeNet5 > KaNet5 by 46.69%	18.01	accepted	F
heatxloop	raw	10000.00	LeNet5 > KaNet5 by 10000.00%	9877.93	not informative	F
heatxloop	norm	-17.70	KaNet5 > LeNet5 by 17.70%	9.16	accepted	F
heatxloop	abs_ordinal	-18.31	KaNet5 > LeNet5 by 18.31%	7.02	accepted	F
heatxloop	zscore	24.41	LeNet5 > KaNet5 by 24.41%	15.26	accepted	F
heatxorl_mask	raw	10000.00	LeNet5 > KaNet5 by 10000.00%	9853.52	accepted	F
heatxorl_mask	norm	-9.61	KaNet5 > LeNet5 by 9.61%	8.54	accepted	F
heatxorl_mask	abs_ordinal	-2.59	KaNet5 > LeNet5 by 2.59%	5.95	accepted	T
heatxorl_mask	zscore	38.45	LeNet5 > KaNet5 by 38.45%	10.68	accepted	F
heatxxvcr_mask	raw	800.78	LeNet5 > KaNet5 by 800.78%	4290.77	accepted	F
heatxxvcr_mask	norm	-16.17	KaNet5 > LeNet5 by 16.17%	5.04	accepted	F
heatxxvcr_mask	abs_ordinal	1.91	LeNet5 > KaNet5 by 1.91%	3.36	accepted	T
heatxxvcr_mask	zscore	50.05	LeNet5 > KaNet5 by 50.05%	21.51	accepted	F
segmentation_heat_loop	raw	0.00	LeNet5 = KaNet5	0.00	accepted	T
segmentation_heat_loop	norm	-0.01	KaNet5 > LeNet5 by 0.01%	0.00	accepted	F
segmentation_heat_loop	abs_ordinal	0.00	LeNet5 = KaNet5	0.00	not informative	F
segmentation_heat_loop	zscore	0.00	LeNet5 = KaNet5	0.00	not informative	F

Table 2: Grad-CAM Metrics: Standard-LeNet5 vs. KaNet5

Metric	Norm. type	$\Delta@50$ (%)	Comparison@50	Δ_{range}	Sign.	Changed
heatxedge	raw	-29.91	KaNet5 > LeNet5 by 29.91%	19.23	accepted	F
heatxedge	norm	-29.91	KaNet5 > LeNet5 by 29.91%	19.23	accepted	F
heatxedge	abs_ordinal	-0.31	KaNet5 > LeNet5 by 0.31%	5.95	accepted	T
heatxedge	zscore	-29.91	KaNet5 > LeNet5 by 29.91%	19.23	accepted	F
heatxcorner	raw	-29.30	KaNet5 > LeNet5 by 29.30%	19.23	accepted	F
heatxcorner	norm	-29.30	KaNet5 > LeNet5 by 29.30%	19.53	accepted	F
heatxcorner	abs_ordinal	-0.31	KaNet5 > LeNet5 by 0.31%	5.65	accepted	T
heatxcorner	zscore	-29.30	KaNet5 > LeNet5 by 29.30%	19.23	accepted	F
heatxloop	raw	-42.72	KaNet5 > LeNet5 by 42.72%	20.75	accepted	F
heatxloop	norm	-42.72	KaNet5 > LeNet5 by 42.72%	20.75	accepted	F
heatxloop	abs_ordinal	-14.65	KaNet5 > LeNet5 by 14.65%	10.99	accepted	F
heatxloop	zscore	-42.72	KaNet5 > LeNet5 by 42.72%	20.75	accepted	F
heatxorl_mask	raw	-31.13	KaNet5 > LeNet5 by 31.13%	23.80	accepted	F
heatxorl_mask	norm	-31.13	KaNet5 > LeNet5 by 31.13%	23.80	accepted	F
heatxorl_mask	abs_ordinal	-1.22	KaNet5 > LeNet5 by 1.22%	9.77	accepted	F
heatxorl_mask	zscore	-31.13	KaNet5 > LeNet5 by 31.13%	23.80	accepted	F
heatxvcr_mask	raw	-29.60	KaNet5 > LeNet5 by 29.60%	18.62	accepted	F
heatxvcr_mask	norm	-29.60	KaNet5 > LeNet5 by 29.60%	18.92	accepted	F
heatxvcr_mask	abs_ordinal	-0.31	KaNet5 > LeNet5 by 0.31%	4.73	accepted	T
heatxvcr_mask	zscore	-29.60	KaNet5 > LeNet5 by 29.60%	18.62	accepted	F
segmentation_heat_loop	raw	-20.75	KaNet5 > LeNet5 by 20.75%	20.75	accepted	F
segmentation_heat_loop	norm	-20.75	KaNet5 > LeNet5 by 20.75%	20.75	accepted	F
segmentation_heat_loop	abs_ordinal	-3.66	KaNet5 > LeNet5 by 3.66%	1.22	accepted	F
segmentation_heat_loop	zscore	-20.75	KaNet5 > LeNet5 by 20.75%	20.75	accepted	F

- **raw**: identity,

$$H_{\text{raw}} = H.$$

Preserves both sign and magnitude of the activations, making it possible to see directly which pixels drive excitatory (positive) or inhibitory (negative) responses.

- **norm** ($[0, 1]$ min-max):

$$H_{\text{norm}} = \frac{H - \min H}{\max H - \min H}.$$

Emphasizes relative activation strengths on a common scale, allowing direct visual comparison of where the network focuses, regardless of absolute intensity.

- **abs_ordinal**: rank order of the absolute values,

$$H_{\text{ord}}(i, j) = \text{rank}(|H_{ij}|) \quad (\text{over all pixels}).$$

Highlights the most salient pixels by ordering them, while ignoring sign, to reveal which regions are deemed most informative.

- **zscore** (abs-z-min-max):

$$\begin{aligned} 1. \quad & A = |H|, \quad Z = \frac{A - \mu(A)}{\sigma(A)}; \\ 2. \quad & H_z = \frac{Z - \min Z}{\max Z - \min Z}. \end{aligned}$$

Filters out outliers and centers activations around their mean, then rescales to $[0, 1]$, exposing pixels that deviate most from average activation.

Score Computation

All six pixel-wise scores combine a normalized heat map H^* with a structural mask or filter extracted from the original image. Let M be the binary or continuous mask of interest; then each score is

$$S = \frac{1}{N} \sum_{i,j} H_{ij}^* M_{ij},$$

except segmentation which counts clusters:

$$S_{\text{segloop}} = |\{\text{unique clusters in quantile-segmented } (H^* \odot M_{\text{loop}})\}|.$$

HeatxEdge $M_{\text{edge}} = |\nabla_x I| + |\nabla_y I|$ (Sobel edges). Measures reliance on object contours.

HeatxCORNER M_{corner} from `cornerMinEigenVal`. Reveals focus on geometric corners.

HeatxLoop M_{loop} from hierarchical contour holes. Captures attention to enclosed regions and cavities.

HeatxORL_Mask M_{orl} from convexity defect opening. Indicates use of concave shape features.

HeatxVCR_Mask M_{vcr} from vertical-run concentration (top 10% columns). Highlights vertical structure sensitivity.

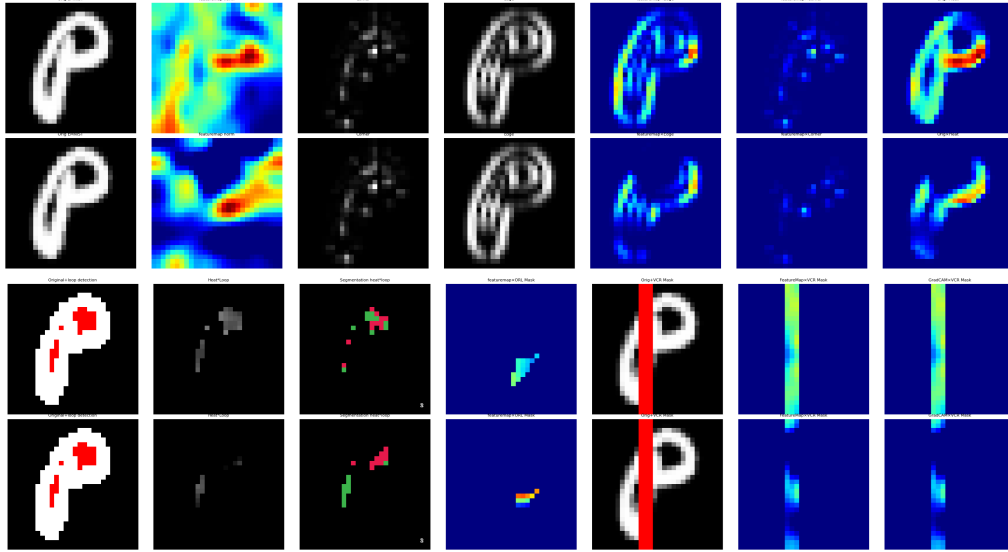
Segmentation_heat_loop Counts clusters in $H^* \odot M_{\text{loop}}$ after 10-quantile segmentation:

$$S = \#\{1, \dots, 10 \text{ clusters with nonzero mask}\}.$$

Quantifies granularity of activation patterns inside loops.

In practice the code:

1. Loads each upsampled heat map H .
2. Applies one chosen transform (`raw`, `norm`, `abs_ordinal` or `zscore`).
3. Extracts masks $\{M_{\text{edge}}, M_{\text{corner}}, \dots\}$ from the original EMNIST image.
4. Computes each score S via element-wise multiplication and averaging (or clustering).
5. Saves the results for statistical comparison (paired t -test) between Standard-LeNet5 and KaNet5.



Comparison grid (4×7) of Std_LeNet5 vs. KaNet5 feature maps / Grad-CAM overlays

Legend of the 4×7 Comparison Grid

- **Row 1 (Std_LeNet5):**
 - Col 1: Orig EMNIST
 - Col 2: FeatureMap norm
 - Col 3: Corner
 - Col 4: Edge
 - Col 5: FeatureMap×Edge
 - Col 6: FeatureMap×Corner
 - Col 7: Orig+loop
- **Row 2 (KaNet5):**
 - Col 1: Orig EMNIST
 - Col 2: FeatureMap norm
 - Col 3: Corner
 - Col 4: Edge
 - Col 5: FeatureMap×Edge
 - Col 6: FeatureMap×Corner

- Col 7: Orig+loop
- Row 3 (Std_LeNet5):
 - Col 1: Original+loop detection
 - Col 2: FeatureMap $\times$ Loop
 - Col 3: Segmentation_heat_loop
 - Col 4: FeatureMap $\times$ ORL_Mask
 - Col 5: Orig+VCR Mask
 - Col 6: FeatureMap $\times$ VCR_Mask
 - Col 7: GradCAM $\times$ VCR_Mask
- Row 4 (KaNet5):
 - Col 1: Original+loop detection
 - Col 2: FeatureMap $\times$ Loop
 - Col 3: Segmentation_heat_loop
 - Col 4: FeatureMap $\times$ ORL_Mask
 - Col 5: Orig+VCR Mask
 - Col 6: FeatureMap $\times$ VCR_Mask
 - Col 7: GradCAM $\times$ VCR_Mask